Problem 1 – Linear Regression, a simple DL (20 Points)

One of the simplest yet useful ML model is the linear regression model where the goal is to find w in order to fit a linear model to the data $z_1, \ldots, z_N$, $z_i = (x_i, y_i)$, where $x_i \in \mathbb{R}^d$ and $y_i \in \mathbb{R}$ are the features and predictions, respectively. Specifically, we aim to solve

$$\min_w \mathcal{L}(w) := \frac{1}{2N} \sum_{i=1}^{N} (y_i - w^\top x_i)^2.$$

a) Show that if $N \geq d$ and $X^\top X$ is invertable, the optimal solution is unique and can be found in closed-form via

$$w^* = (X^\top X)^{-1} X^\top y,$$

where $y = [y_1, \ldots, y_N]^\top \in \mathbb{R}^N$ and $X = [x_1, \ldots, x_N]^\top \in \mathbb{R}^{N \times d}$.

b) Assume instead of using the closed-form solution, we aim to solve the linear regression problem via SGD. Write down the SGD update rule explicitly for this task.

c) Let $d = 10$ and $N = 100$, and $\sigma = 0.01$. Write a Python program to solve a linear regression problem using SGD with $T = 2000$ iterations and a constant learning rate. You may generate the data such that each entries of x_i and w^* are drawn from standard normal distribution. Plot the linear regression loss function $\mathcal{L}(w_t)$ versus $t = 1, \ldots, T$ and report the learning rate you used. Submit your code as a PDF file.

d) Repeat the above experiment now with $d = 200$ instead and report your findings.

Problem 2 – Predictive Gradient Descent I (30 Points)

Consider the task of finding w^*, a global minimum of the convex function $\mathcal{L}(w)$, $w \in \mathbb{R}^d$. To this end, instead of using Gradient Descent (GD), we employ the following variant:

$$\begin{aligned} x_{t+1} &= x_t - \eta \nabla \mathcal{L}(w_t) \\ w_{t+1} &= x_{t+1} - \eta \mathcal{P}_{t+1}, \end{aligned} \tag{1}$$

where $x_1 = w_1$ are the initialization points, η is a learning rate, T is the number of iterations, and $\mathcal{P}_t,\ t = 1, \dots T$ is a sequence of d-dimensional vectors given to us as the inputs of the algorithm. Let us call this algorithm Predictive Gradient Descent (PGD).

(a) Show that

$$\mathcal{L}(w_t) - \mathcal{L}(w^*) \leq \langle \nabla \mathcal{L}(w_t), x_{t+1} - w^* \rangle + \langle \mathcal{P}_t, w_t - x_{t+1} \rangle + \langle \nabla \mathcal{L}(w_t) - \mathcal{P}_t, w_t - x_{t+1} \rangle. \tag{2}$$

(b) Show that

- $\langle \nabla \mathcal{L}(w_t), x_{t+1} - w^* \rangle = \frac{1}{2\eta} \Big[\|x_t - w^*\|^2 - \|x_t - x_{t+1}\|^2 - \|x_{t+1} - w^*\|^2 \Big]$,
- $\langle \mathcal{P}_t, w_t - x_{t+1} \rangle = \frac{1}{2\eta} \Big[\|x_t - x_{t+1}\|^2 - \|x_t - w_t\|^2 - \|w_t - x_{t+1}\|^2 \Big]$,
- $\langle \nabla \mathcal{L}(w_t) - \mathcal{P}_t, w_t - x_{t+1} \rangle \leq \frac{1}{2\eta} \|w_t - x_{t+1}\|^2 + \frac{\eta}{2} \|\nabla \mathcal{L}(w_t) - \mathcal{P}_t\|^2$.

(c) Using the previous part show that there exists a $\hat{w}$ such that

$$\mathcal{L}(\hat{w}) - \mathcal{L}(w^*) \leq \frac{1}{2\eta T} D^2 + \frac{\eta}{2T} \sum_{t=1}^{T} \|\nabla \mathcal{L}(w_t) - \mathcal{P}_t\|^2, \tag{3}$$

where $D = \|x_1 - w^*\|$.

Problem 3 – Predictive Gradient Descent II (35 Points)

Consider the function $\mathcal{L}(w) = w(w^2/3 - \bar{w}^2)$, $w \in \mathbb{R}$, on the interval $[-\sqrt{3}\bar{w}, \sqrt{3}\bar{w}]$.

(a) Prove or disprove whether

- $\mathcal{L}(w)$ is convex,
- $\mathcal{L}(w)$ is G-Lipschitz for some $G > 0$,
- $\mathcal{L}(w)$ is L-smooth for some $L > 0$.

(b) Find the global minimizer of $\mathcal{L}(w)$ on the interval $[-\sqrt{3}\bar{w}, \sqrt{3}\bar{w}]$. That is, find

$$w^* = \underset{w \in [-\sqrt{3}\bar{w}, \sqrt{3}\bar{w}]}{\arg\min} \mathcal{L}(w).$$

(c) Let $\bar{w} = 1$. Consider the *Projected Gradient Descent (PGD)* Algorithm to find w^* with the following update:

$$w_{t+1} = \Pi\Big[w_t - \eta \nabla \mathcal{L}(w_t)\Big],$$

where w_1 is the initialization, η is the learning rate, and

$$\Pi(x) = \begin{cases} x & |x| \leq \sqrt{3} \\ \sqrt{3} & x > \sqrt{3} \\ -\sqrt{3} & x < -\sqrt{3}. \end{cases}$$

Write a Python program and implement PGD with $w_1 = -0.9$ and $\eta = \frac{1}{2\sqrt{3}}$ to minimize $\mathcal{L}(w)$. Submit your code as an attachment to your solution.

- How many iterations does it take for PGD to converge?
- Does PGD converge to w^*?
- Repeat the experiment with $w_1 = -1.1$ and $w_1 = -1$ and again answer the previous questions.

(d) Assume instead of PGD, we are interested in the following update

$$w_2 = w_1 - \eta \nabla \mathcal{L}(w_2). \tag{4}$$

Notice that the gradient is w.r.t. w_2 not w_1. Show that as long as

$$w_1 > -\Big(\frac{1}{4\eta} + \eta \bar{w}^2\Big),$$

w_2 has a closed form expression as a function of w_1, η, and $\bar{w}$.

(e) Find the limit of w_2 and the feasible range of w_1 as $\eta \to \infty$.

Problem 4 – ZO-SGD with one point estimator (15 Points)

As we discussed in class, for a function ℓ defined on $\mathbb{R}$,

$$g^u(w) = \frac{\ell(w + \mu u) - \ell(w)}{\mu} u,$$

where u is drawn uniformly from the unit ball, is an unbiased estimator of the gradient of $\ell_\mu(w) = \mathbb{E}_u[\ell(w + \mu u)]$, an approximation of the derivative of ℓ. Using a similar technique, argue that

$$\rho^u(w) = \frac{\ell(w + \mu u)}{\mu} u,$$

is also an unbiased estimator of the gradient of $\ell_\mu(w) = \mathbb{E}_u[\ell(w + \mu u)]$, an approximation of the derivative of ℓ (Hint: you may find the expectation of ρ^u and show that it is equal to the expectation of g^u when u is sampled uniformly from the unit ball in $\mathbb{R}$).